

Results

01 • NESTED ANOVA

one factor nested within another

Table 1. Descriptives by top-level group

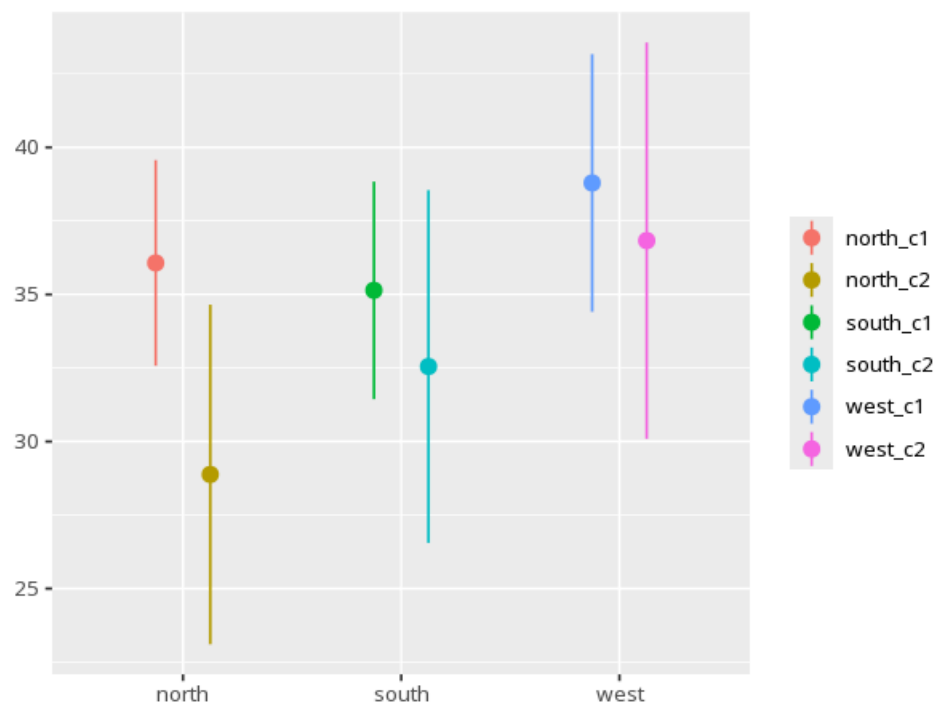
Group	N	M	SD
north	20	32.48	7.47
south	20	33.84	6.91
west	20	37.81	7.79

Table. Nested ANOVA

Source	SS	df	MS	F	p	ω^2 [95% CI]
school	307.07	2	153.53	1.48	.357	0.06 [0.00, 1.00]
classroom (nested in school)	311.23	3	103.74	1.99	.126	0.04 [0.00, 1.00]

Under random nesting the F for the upper factor (A) uses the nested factor's mean square B(A) as its error term, while B(A) is tested against the residual — so the two F rows do not share the same denominator. Variance components (or ω^2) are reported as the effect size where estimable. Assumption checks: Levene's (equal variances across top-level groups) & normality of residuals (Shapiro-Wilk). (Levene $F=0.02$, $p=.978$ · Shapiro $W=0.98$, $p=.297$)

Figure. Grouped means



How to read this test

Used when one factor sits inside another (e.g. classes within schools). The top factor is tested against variation among its nested units, not raw residuals — a significant F means the top-level groups differ beyond the nested-unit variability. Your significance threshold (α) is 0.05.

APA template: A nested ANOVA for A gave $F(2,3)=1.48$, $p = .357$, $\omega^2=.06$ [.00, 1.00].

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